

STOCKDOCK: AN ENTERPRISE POS & MULTI-BRAND SUPPLY CHAIN CLOUD MANAGEMENT SYSTEM

1. Introduction

In traditional Fast-Moving Consumer Goods (FMCG) and perishable food distribution networks such as the daily bakery and bread supply chain in urban India retail store owners and regional distributors face severe operational inefficiencies. Daily inventory tracking is predominantly manual, leading to inaccurate reordering, overstocking, and substantial financial losses due to unsold expired inventory. Furthermore, handling orders across multiple independent commercial brands (e.g., Relish, English Oven, Max Heath) creates fragmented billing and administrative friction.

StockDock is a cloud-native, enterprise-grade Point of Sale (POS) and Supply Chain Management platform engineered to bridge local retail store owners and regional distributors through a unified web application. Built on a modern React, Node.js, Express, and MongoDB architecture, StockDock automates daily stock matrix calculations, provides multi-server bank failover UPI payment routing, and equips distributors with company-wise financial reporting and CSV data export capabilities.

2. Objectives

- **Digitize Daily Inventory Matrices:** Automate stock replenishment calculations ($\text{Load} = \text{Target Stock} - (\text{Current Left} - \text{Expired})$) across multi-brand catalogs without requiring manual paper logging.
- **Implement a Two-Step Linear Checkout & Tax Invoice Engine:** Provide store owners with itemized tax invoice generation, instant printable PDF rendering, and standalone checkout routing.
- **Provide Multi-Server UPI Failover Payment Gateways:** Mitigate single-bank server downtime during digital settlements by providing real-time routing across 3 primary/backup banking accounts (HDFC, ICICI, and SBI) with dynamic SVG QR code generation and 10-minute session security timers.
- **Deliver Advanced Distributor Business Intelligence:** Enable distributors to monitor network-wide store performance, view company-wise revenue breakdowns, filter delivery histories by daily/monthly periods, and safely manage records with automated CSV download fallbacks.

4. Stakeholders

Stakeholders represent the key entities interacting with and benefiting from the system:

- **1. Client (Enterprise / Regional Logistics Agencies):**
 - The primary operating authority managing the distribution network, establishing merchant UPI endpoints, and overseeing multi-brand sales.

- **2. Retail Store Owners (End Users - Retail):**

- **Role / Designation:** Local Retailers, Bakery Owners, POS Operators.
- **Actions Performed:**
 - Log in to access a centralized Store Management Center.
 - Browse multi-brand product catalogs (Relish, English Oven, Max Heath).
 - Input daily stock levels ("Current Left" and "Expired") to auto-calculate the daily required cargo.
 - Generate official master tax invoices and inspect itemized totals.
 - Settle unpaid inventory bills via multi-bank UPI QR codes or view past electronic payment receipts.

- **3. Regional Distributors (End Users - Logistics):**

- **Role / Designation:** Supply Chain Managers, Agency Logistics Directors.
- **Actions Performed:**
 - Monitor store-by-store and network-wide sales vs. expiry analytics via interactive Recharts bar graphs.
 - Inspect live cargo dispatch notices and individual shop delivery breakdowns.
 - Filter delivery history by daily, monthly, or custom timeframes.
 - Generate aggregated financial sales reports and export company-wise revenue data to CSV spreadsheets.
 - Delete obsolete dispatch records with a mandatory "Download CSV & Delete" safeguard.

- **4. Technical User / System Administrator:**

- **Role / Designation:** Software Developer, System Admin.
- **Actions Performed:** Database maintenance on MongoDB, JWT secret management, and REST API endpoint maintenance.

5. Expected Modules and Sub-modules

Module 1: Authentication & Role-Based Access Control (RBAC)

- **Register Sub-Module:** Allows new enterprise users to register as either a "Store Owner" (with shop name, area address, mobile number, and operating hours) or a "Distributor" (with agency name and warehouse address).

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- **Sign-In Sub-Module:** Authenticates users using Bcrypt password hashing and issues a 2-hour JSON Web Token (JWT) for secure session persistence.
- **Session & Navigation Sub-Module:** Automatically routes authenticated users to their respective role-specific dashboard and preserves user preferences (language and theme).

Module 2: Store Owner Order Catalog & Inventory Management

- **Multi-Brand Catalog Browser:** Displays master inventory items organized under brand tabs (*All Brands, Relish, English Oven, Max Heath*) with unit prices.
- **Auto-Calculating Daily Stock Matrix:** Automatically calculates exact billable units and required cargo loads dynamically without requiring manual checkbox toggles, filtering out zero-movement stock automatically.

Module 3: Two-Step Checkout & UPI Payment Gateway

- **Tax Invoice Review Sub-Module:** Displays a dedicated, printable tax invoice showing reference numbers, billed shop details, itemized breakdown, and master total due.
- **Multi-Bank Failover Routing Sub-Module:** Allows switching across 3 bank servers (HDFC Primary, ICICI Backup, SBI Backup) to ensure 100% payment uptime if a specific bank gateway is down.
- **Dynamic SVG QR & Session Timer Sub-Module:** Renders a real-time `upi://pay` URI SVG QR code with a 10-minute security countdown timer.
- **1-Click Manual Settlement Sub-Module:** Updates payment status to PAID in the database and generates an electronic receipt with transaction ID and timestamp.

Module 4: Distributor Network Analytics & Delivery History

- **Network Performance Visualizer:** Interactive Recharts bar graph comparing total items sold vs. expired returns across individual stores or network-wide totals.
- **Delivery Dispatch History:** Comprehensive ledger of all dispatched cargo orders with daily/monthly time filtering and invoice modal viewers.
- **Safeguarded Record Deletion:** Modal interface requiring distributors to choose between immediate deletion or a 1-click "Download CSV & Delete" backup.

Module 5: Financial Reporting & CSV Export Engine

- **Company-Wise Revenue Aggregator:** Calculates and displays revenue generated by each brand (e.g., Relish vs. English Oven vs. Max Heath) for selected timeframes.
- **Sales Trend Visualizer:** Dual-axis line chart illustrating sales volume vs. total revenue trends.

- **One-Click CSV Exporter:** Exports complete aggregated financial reports directly into formatted CSV files for accounting software integration.

6. Hardware and Software Requirements

Hardware Requirements

- **Minimum Processor:** Dual-Core Intel Core i3 / AMD Ryzen 3 (2.0 GHz or higher)
- **Primary Memory (RAM):** 4 GB Minimum (8 GB Recommended)
- **Secondary Storage:** 256 GB Solid State Drive (SSD)
- **System Architecture:** 64-bit x86 Operating Architecture

Software Requirements

- **Operating System:** Windows 10 / 11, macOS, or Linux (Ubuntu 20.04+)
- **Programming Languages:** JavaScript (ES6+), JSX
- **Frontend Framework & Libraries:**
 - React.js 18
 - Vite (Build Tool / Server)
 - Tailwind CSS (v4 Styling Framework)
 - Lucide React (Vector Icon Library)
 - Recharts (Data Visualization)
 - qrcode.react (SVG UPI QR Code Generation)
 - Axios (HTTP Client)
- **Backend Runtime & Framework:**
 - Node.js (v18.0.0 or higher)
 - Express.js (REST API Web Framework)
 - JSON Web Token (JWT Authentication)
 - BcryptJS (Password Hashing)
- **Database & ORM:**
 - MongoDB Community Server (v6.0+)
 - Mongoose ORM
 - MongoDB Compass (Database GUI Manager)
- **Network & Protocols:**

- Network Environment: Localhost / Local Area Network / Web Cloud
- Protocols: HTTP/1.1, REST API over TCP/IP
- Ports Used: Frontend (Port 5173), Backend Server (Port 5005), MongoDB (Port 27017)

7. Similar Software Study / Background Study

Title

Vyapar App / Khatabook

Type

Desktop & Mobile SME Accounting, POS, and Billing Application.

Hardware and Software Requirements

- **Operating System:** Android 7.0+ / Windows 10+
- **Hardware:** 2 GB RAM, 100 MB Storage
- **Software Stack:** Native Android / Electron, SQLite / Proprietary Cloud Sync

Modules

- Billing & Tax Invoicing
- Inventory & Stock Tracking
- Customer/Supplier Ledger Management
- Payment Reminder Engine (WhatsApp/SMS)

Salient Features

- Simple offline-first billing UI for small kirana stores.
- WhatsApp PDF invoice sharing.
- Basic inventory quantity tracking and GST tax computation.

Limitations

- **No Automated Perishable Matrix Logic:** Requires users to manually enter every line item and calculation without automatic generation of good stock vs. expired returns logic.
- **No Multi-Server Bank Failover:** Uses static payment links without dynamic UPI multi-account routing or bank failure switching natively built into the checkout process.

- **Single-Sided Architecture:** Designed primarily for standalone retailers rather than establishing a real-time, dual-sided ecosystem connecting retail store owners directly with multi-brand regional distributors.

Remark

While Vyapar and Khatabook serve as general-purpose SME accounting tools, **StockDock** is specialized for FMCG and perishable supply chains. By pairing an automated daily stock matrix with multi-server payment failover and distributor-level multi-brand reporting, StockDock delivers a complete end-to-end supply chain cloud platform.